SUBIN SIMON

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PROFILE

MTech artificial intelligence student with hands-on experience in **deep learning, computer vision, time-series forecasting**, and cloud-based deployment. Proficient in Python, TensorFlow, PyTorch, and Scikit-learn with strong foundations in data pre-processing, model evaluation, and neural network design. Experienced in developing end-to-end ML solutions through academic and project-based research, with interests in AI-driven prediction systems and intelligent applications.

EDUCATION

Master of Technology in Artificial Intelligence 17/2024 – present
Amrita Vishwa Vidyapeetham Coimbatore, India
CGPA: 7.2

Bachelor of Technology in Computer Science and Engineering 06/2018 – 06/2022
Karunya Institute of Technology and Sciences Coimbatore, India
CGPA : 8.52

Higher Secondary Education 05/2017 – 05/2018
Kendriya Vidyalaya Trivandrum, India
Score : 80.8%

SSLC 05/2015 – 05/2016
Kendriya Vidalaya Trivandrum, India
CGPA : 10

SKILLS

Languages: C / C++, Python, HTML & CSS, SQL

ML & AI: CNN, LSTM, Time series forecasting, Computer vision

Cloud: AWS (EC2, S3, Lambda)

Libraries & Frameworks: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Mamba-ssm

Soft Skills: Communication, Leadership, Team collaboration, Adaptability

PROJECTS

Spatio-Temporal Monsoon Change Prediction
Tech Stack: Python, PyTorch, Mamba-SSM
- Developed a structured state-space deep learning framework (**Mamba architecture**) for regional monsoon rainfall severity prediction (1995–2025 ERA5-Land dataset).
- Aggregated hourly meteorological data to monthly resolution and engineered temporal features for seasonal modeling.
- Achieved 88.9% classification accuracy and macro F1-score of 0.64 under class imbalance conditions.
- Implemented confusion matrix analysis and macro-metric evaluation for robust performance assessment.
- Built user-defined forecasting module for dynamic rainfall severity prediction.

Emotion Detection For Depressed Teenagers.
Tech Stack: Python, TensorFlow/Keras, FER-2013 Dataset, CNN
• Developed a convolutional neural network (CNN) model for facial emotion recognition using the FER-2013 dataset.
• Performed data preprocessing, augmentation, and normalization to improve generalization.
• Classified emotions such as sadness, anger, and disgust to support early mental health assessment.
• Improved robustness of emotion detection by eliminating manual feature engineering.

Student Mark Prediction Using Machine Learning Techniques
Tech stack: Python.Scikit-learn, Pandas, Flask.
• Built machine learning models to predict student academic performance based on behavioral and demographic features.
• Conducted data preprocessing, exploratory data analysis (EDA), and model evaluation using Scikit-learn.
• Deployed the trained model as a web application using Flask for real-time prediction.

Infinity - An NFT Marketplace On Ethereum
Tech stack: Ethereum

- Developed a decentralized NFT marketplace allowing users to list,buy and sell digital assests.
- Implemented secure ownership transfer through smart contracts, handling transactions between buyers and sellers.
- Enabled revenue generation via listing fees, contributing to sustainable marketplace operations.
- Ensured seamless asset listings and transaction process using ethereum blockchain.

CERTIFICATES

Introduction to Large Language Models (NPTEL)

Responsible & Safe AI Systems (NPTEL)

Applied Social Network Analysis in Python
- Represent and manipulate networked data using the NetworkX library.
- Predict the evolution of networks over time.

India_Internship_Karunya University_Introduction to Cybersecurity

Cambridge English Level1, ESOL International (Business Vantage)

LANGUAGES

- English
- Hindi
- Malayalam

HOBBIES

Cricket

Football

Athletics

ACKNOWLWDGMENT

I hereby declare that all the details provided above are true to my knowledge.

Subin Simon